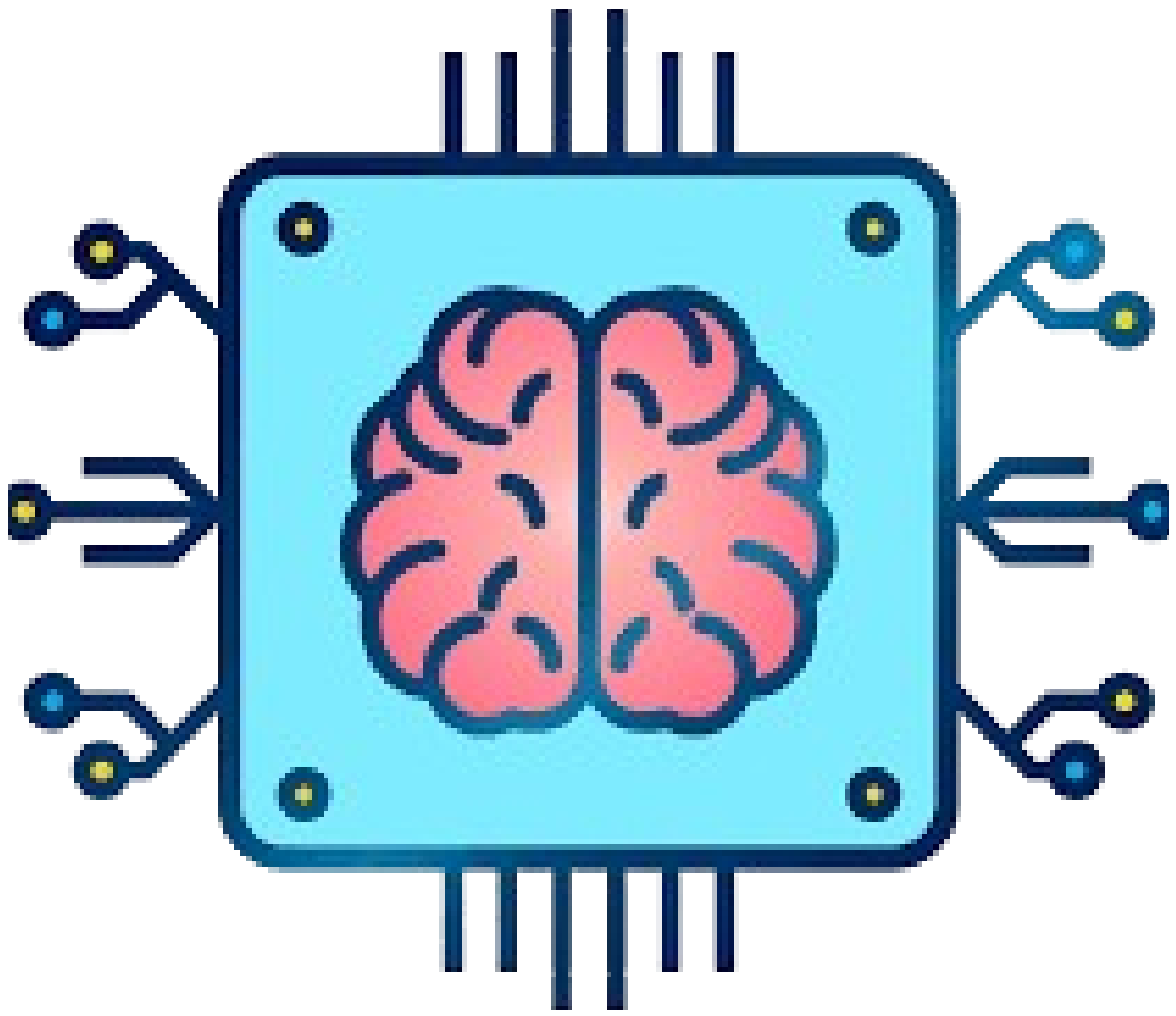


Agentic AI

Cheat Sheet



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AI Engineer

AI Agents

Systems that actautonomously to achieve goals using reasoning, memory, and tools.

Built on top of LLMs (e.g., GPT-4o, Claude, Gemini)with planning + execution loops.

Types

Reactive Agents–respond instantly (e.g., chatbots).

Planning Agents – reason, plan, and execute multi-step tasks.

Collaborative Agents – coordinate with other agents or humans.

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Agent Frameworks

Framework	Description	Highlights
LangChain	Modular framework for reasoning + tool use	Chains, Agents, Memory
LlamaIndex	Data indexing + retrieval for LLMs	RAG, Context management
CrewAI / AutoGen	Multi-agent collaboration frameworks	Role-based AI teamwork
OpenDevin / BabyAGI	Autonomous coding + task agents	Self-improving workflows
Microsoft Semantic Kernel	Enterprise-grade agent orchestration	C#, Python SDKs
AutoGPT / SuperAGI	Open-source autonomous agents	Tool use + long-term memory

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Agent Architecture

Planner → decides next actions

Executor → performs actions via tools or APIs

Memory → short-term (chat context), long-term (vector store)

Reasoner → evaluates progress and adapts plan

Interface → connects with user or system

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Key Components

LLM Backbone → GPT-4o, Claude 3, Gemini 1.5

Vector DBs → Pinecone, FAISS, Chroma, Weaviate

Tools → Google Search API, Python REPL, Zapier, SQL Connector

Memory Systems → LangChain Memory, Redis, Milvus

Orchestration → ReAct, Plan-Act- Reflect, AutoGen loops

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Agentic Patterns

ReAct Pattern:
reasoning + acting alternately

Plan-Execute Pattern:
plan → execute → refine

Reflection Loop:
agent reviews and self-corrects

Team Agent Pattern:
multiple agents with roles (e.g., planner, coder, reviewer)

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Evaluation Metrics

Area	Metric	Description
Reasoning	Task Success Rate	Can it reach goal autonomously?
Planning	Step Efficiency	Fewer, correct steps
Execution	Tool Accuracy	Correct use of APIs/tools
Safety	Constraint Adherence	No harmful or disallowed actions

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Agent Use Cases

Business Automation: Email, Reporting, Scheduling

Data Analysis: Querying, Insights, Dashboards

Code Generation: Debugging, Testing, Deployment

Knowledge Retrieval: RAG, Document Q&A

Creative Workflows: Content, Design, Marketing

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Tools & Platforms

Category	Examples
Frameworks	LangChain, AutoGen, CrewAI
Memory & DB	Chroma, Pinecone, FAISS
Vector Search	Weaviate, Milvus
Evaluation	TruLens, LangSmith
Hosting	OpenAI API, Hugging Face, Azure AI Studio

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